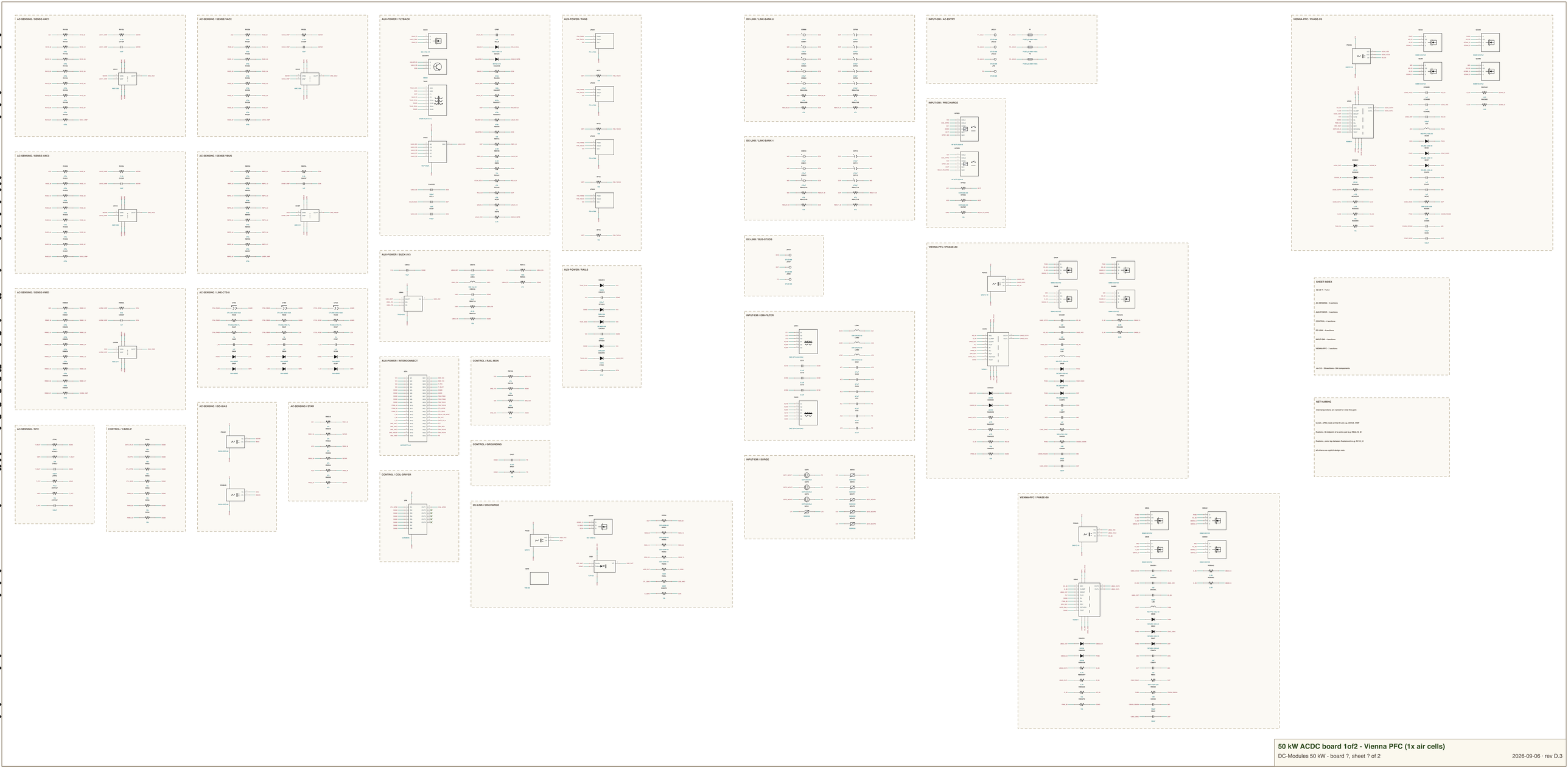
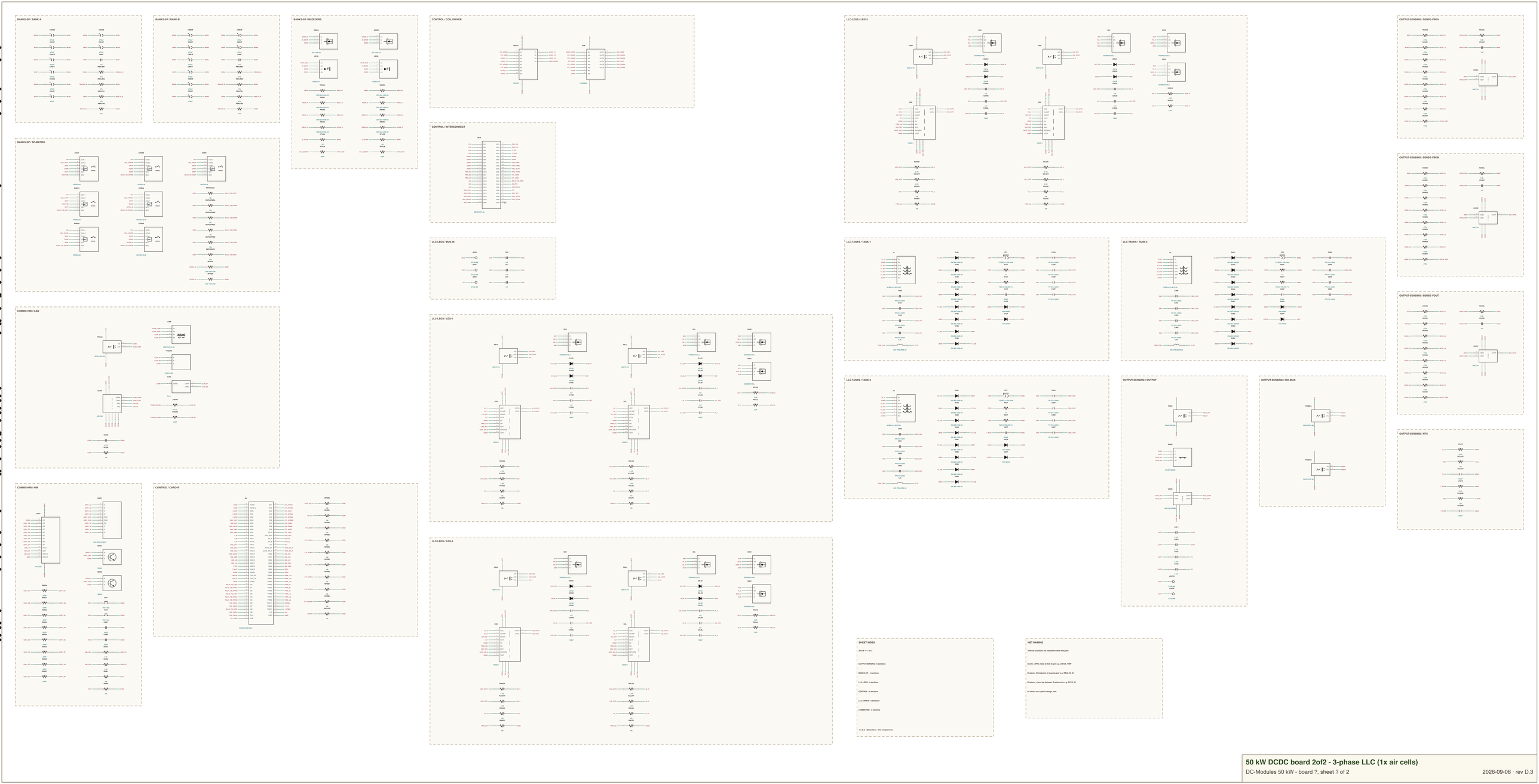
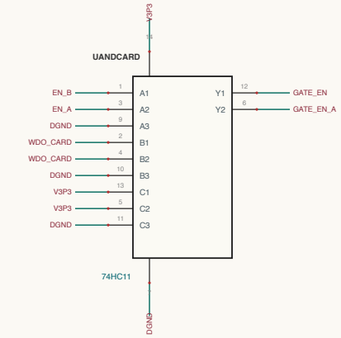
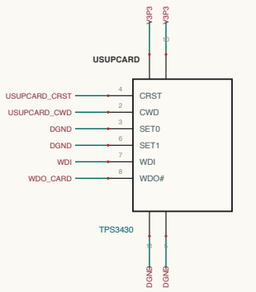

AC-SENSING / SENSE-VAC1
AC-SENSING / SENSE-VAC2
AUX-POWER / FLYBACK
AUX-POWER / FANS
DC-LINK / LINK-BANK-G
INPUT-EMI / AC-ENTRY
VIENNA-PFC / PHASE-C0
AC-SENSING / SENSE-VAC3
AC-SENSING / SENSE-VBUS
DC-LINK / LINK-BANK-I
AC-SENSING / SENSE-VMID
AC-SENSING / LINE-CTS-G
AUX-POWER / BUCK-3V3
AUX-POWER / RAILS
DC-LINK / BUS-STUDS
INPUT-EMI / PRECHARGE
VIENNA-PFC / PHASE-A0
AUX-POWER / INTERCONNECT
CONTROL / RAIL-MON
INPUT-EMI / EMI-FILTER
AC-SENSING / NTC
CONTROL / CARD-IF
AC-SENSING / ISO-BIAS
AC-SENSING / STAR
CONTROL / COIL-DRIVER
CONTROL / GROUNDING
DC-LINK / DISCHARGE
INPUT-EMI / SURGE
VIENNA-PFC / PHASE-B0
SHEET INDEX
NET NAMING
50 kW ACDC board 1of2 - Vienna PFC (1x air cells)
DC-Modules 50 kW - board ?, sheet ? of 2
2026-09-06 · rev D.3

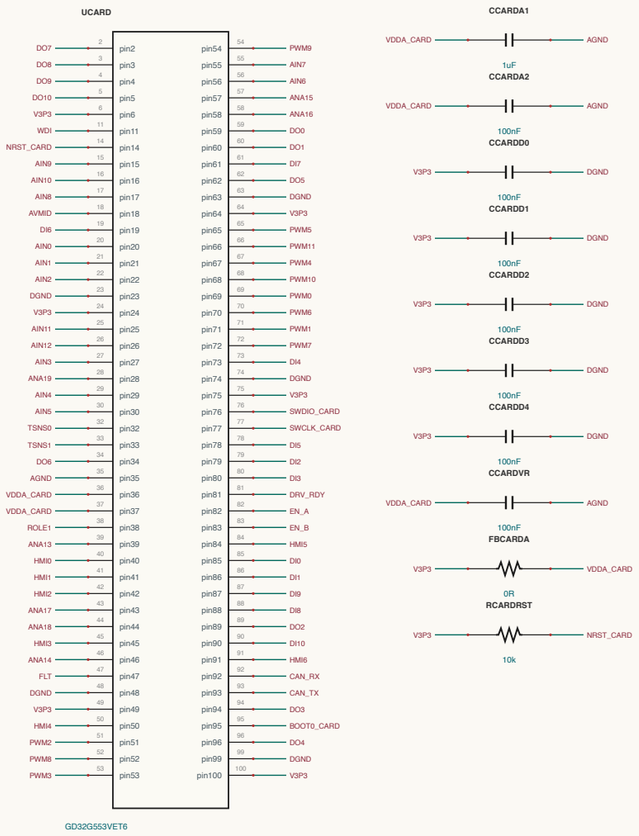

BANKS-SP / BANK-A
BANKS-SP / BANK-B
BANKS-SP / BLEEDERS
CONTROL / COIL-DRIVER
CONTROL / INTERCONNECT
LLC-LEGS / LEG-2
OUTPUT-SENSING / SENSE-VBKA
OUTPUT-SENSING / SENSE-VBKB
OUTPUT-SENSING / SENSE-VOUT
BANKS-SP / SP-MATRIX
LLC-LEGS / BUS-IN
LLC-TANKS / TANK-1
LLC-TANKS / TANK-3
COMMS-HMI / CAN
LLC-LEGS / LEG-1
LLC-TANKS / TANK-2
OUTPUT-SENSING / OUTPUT
OUTPUT-SENSING / ISO-BIAS
COMMS-HMI / HMI
CONTROL / CARD-IF
LLC-LEGS / LEG-3
OUTPUT-SENSING / NTC
SHEET INDEX
NET NAMING
50 kW DCDC board 2of2 - 3-phase LLC (1x air cells)
DC-Modules 50 kW - board ?, sheet ? of 2
2026-09-06 · rev D.3

CONTROL / SAFETY

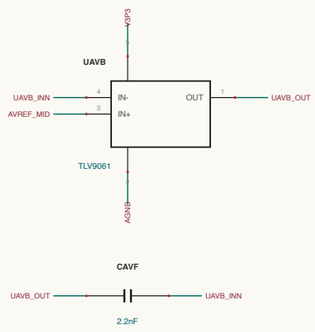


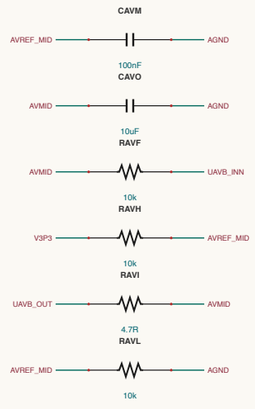


CONTROL / MCU

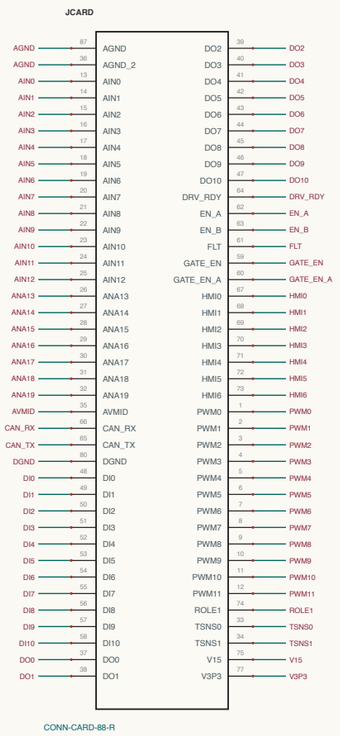


CONTROL / ANALOG-MID

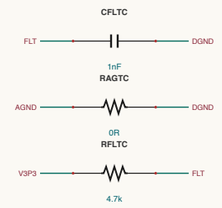




CONTROL / CARD-IF

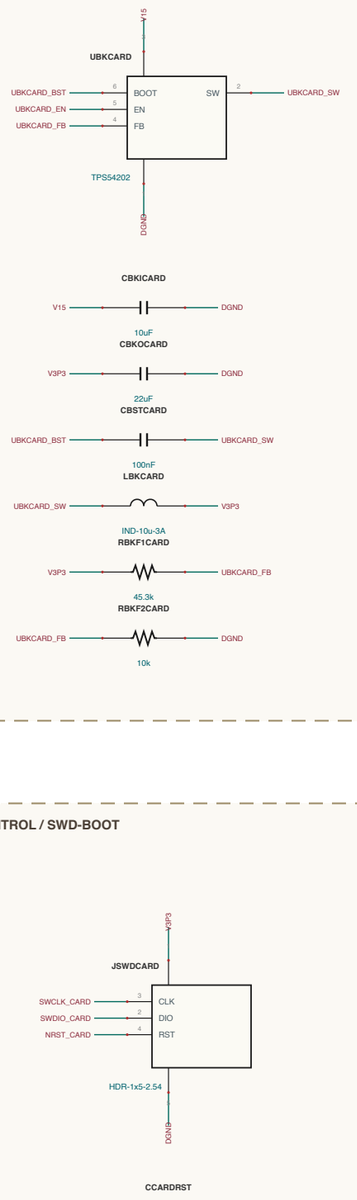


CONTROL / FLT-GROUND

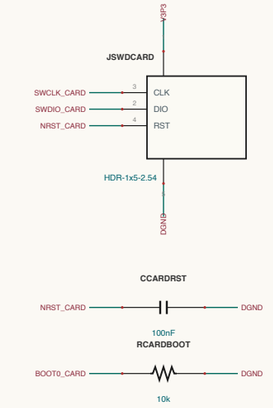


CONTROL / ROLE

CONTROL / BUCK-3V3



CONTROL / SWD-BOOT



30/60 kW Control card - 3 of 3 · rev D.3 · 8 sections · 45 components

NET NAMING: U<ref>_<PIN> = node at that IC pin · R<stem>_M = series-pair midpoint · R<stem>_<nn> = tap between R<stem>H/m · all others are explicit design nets

DC-Modules — Control Card (GD32G553VET6, one card for both converter roles)

DC-Modules 30/60 kW - board Control card, sheet 3 of 3

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